

# Ho Ai Ning (Miriam)

Senior Software Engineer · AI Systems & LLM Integration · Python · Laravel · AWS · Remote (Worldwide)

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## PROFESSIONAL SUMMARY

Senior software engineer building production LLM systems on top of high-scale platforms. Sole owner of property-ai, the natural-language query and conversational layer for a property platform serving 7M+ listings across 111 countries (IQI Global, Juwai.asia) — LLM-based query understanding with a deterministic fallback path, retrieval-grounded chat, and a self-hosted geospatial routing engine deployed on AWS within a fixed infrastructure budget. Python and PHP/Laravel across AI services and the platforms they sit on; React, TypeScript and PostgreSQL on the product side. Independent shipper: took a commercial AI SaaS product from concept to paying beta in under two months alongside full-time work. Writing production code since 2020. Open to senior remote IC roles, worldwide. Multilingual: English, Mandarin, Malay, Cantonese.

## TECHNICAL SKILLS

**AI & LLM Systems:** LLM structured extraction (natural language to typed filters), graceful-degradation design for LLM-dependent paths, RAG pipelines built from scratch (chunking, embeddings, vector similarity, grounded generation — no vector database or orchestration framework), relevance ranking, prompt design and evaluation, OpenAI and Claude APIs, ElevenLabs

**Languages:** Python (FastAPI) · PHP (Laravel) · JavaScript (ES6+) · TypeScript · SQL · Java · HTML/CSS

**Backend & Infrastructure:** FastAPI, REST API design, microservice architecture, PostgreSQL, MySQL, MariaDB, Supabase, AWS (EC2, S3, Lambda), Linux (Ubuntu/CentOS), Git, CI/CD

**Frontend & Mobile:** React, Next.js, Flutter, SvelteKit, Tailwind CSS, Leaflet.js, Laravel Filament

## PROFESSIONAL EXPERIENCE

### Full-Stack Developer (AI Systems) | IQI Global

Dec 2025 – Present

*Sole owner of property-ai — the LLM query-understanding and conversational layer for a property platform serving 7M+ listings across 111 countries on iqiglobal.com and juwai.asia.*

- Architected, deployed and maintain property-ai end-to-end: natural-language property search, conversational assistant, geospatial routing, and multi-domain API integrations — live across 2 production domains and multiple repositories, owned solo from design to on-call.
- Built LLM structured extraction that converts free-text queries ("3-bed condo near KLCC under 800k") into typed search filters against the platform's listing API, replacing rigid form-based search with natural language.
- Designed a deterministic regex fallback so search degrades rather than fails when the LLM provider is unavailable — common constraints (bedrooms, property type, price) still resolve, keeping a core user journey functional during outages of a third-party dependency.
- Built a retrieval-grounded chatbot for platform service questions, implementing the full RAG pipeline from scratch in a legacy PHP stack — document chunking, embeddings, vector storage and similarity matching — with no vector database or orchestration framework. Generation constrained to retrieved content to prevent fabricated service claims.
- Wrote the chatbot relevance ranking that orders the top 5 results by closeness of match to user intent.
- Deployed and maintain a geospatial isochrone engine on self-hosted Valhalla (AWS t3.small) enabling commute-time-based property filtering — a first-of-kind search capability for the platform, running at a fraction of commercial routing-API cost.
- Building a Python XML onboarding engine to automate bulk property-listing ingestion, currently a manual multi-day process; target is a same-day pipeline for global agency partners.
- Led convergence of four legacy domains (juwai.com, juwai.asia, list.juwai.com, m.juwai.com) into a unified architecture, improving SEO and cross-platform data consistency.
- Built an AI podcast generator (Claude API + ElevenLabs) converting property briefing PDFs into two-host audio episodes; deployed and ready for company-wide rollout.
- Implemented a dual-provider map strategy (OSM/Esri) with IP-based switching for optimised global rendering performance.
- Maintained code quality through PR review, automated testing, and scalable schema design in an Agile workflow.

### Founder & Engineer | sanctuary — AI Journaling SaaS

Apr 2026 – Present

*Commercial AI product taken from concept to paying closed beta in under two months, solo, alongside full-time employment.*

- Built and shipped the full stack independently: Flutter (iOS/Android), Supabase (PostgreSQL + auth), Claude API (Haiku/Sonnet) for the AI companion, AdMob, and a Next.js/TypeScript marketing site — running on a ~USD 70/month infrastructure budget.
- Designed the AI companion around contextual reflection rather than generic prompting: conversation state, memory across entries, and model tiering (Haiku/Sonnet) to hold per-user inference cost within unit economics.
- Owned the commercial layer end-to-end: tiered pricing (USD 7.99/mo, USD 49.99/yr), RevenueCat billing, and full legal compliance (Malaysian PDPA, GDPR).

### Freelance Software Engineer | Independent

Sep 2021 – Present

*36+ projects delivered; clients included Maxis, Digi, and TM One.*

- Architected and deployed real-time streaming platforms for Maxis, Digi and TM One (Laravel, PostgreSQL, MariaDB, GitLab CI/CD on Linux), including custom analytics dashboards supporting 200+ concurrent viewers.
- Delivered national-level HR management and permit & licensing platforms (Joget, Java, MySQL).
- Ran full project lifecycles independently: scoping, contracts, delivery, and client relationships.

**Associate Software Developer | Mokxa Technologies LLC (US)** May 2024 – Jun 2025

*Fully remote role with a US employer; delivered enterprise platforms in partnership with Silverlake Innovation Partners and Infosys.*

- Led development of the HRDC National Competency Standard platform (Java + Joget), owning the Event and Course Management modules.
- Built a permit and licensing system deployed across US and Canadian jurisdictions, handling multi-region regulatory workflows.
- Redesigned the employee portal, achieving an 80% reduction in maintenance time through optimised CMS workflows and automated content updates.

**Product Specialist | Joget, Inc.** Oct 2022 – Apr 2024

- Developed and debugged Joget plugins in Java and built internal business applications; authored knowledge-base documentation and produced official platform tutorial content (60,000+ views).

**Junior Full-Stack Developer | Cake Experiential Communications** Jul 2020 – Feb 2021

- Developed real-time streaming platforms for Maxis, Digi and TM One (Laravel, PostgreSQL, MariaDB, Linux).
- Supported disaster recovery implementation for Affin Bank: web application failover and database mirroring across environments.

**CERTIFICATIONS**

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| <b>AWS Certified Cloud Practitioner</b> | 2026 |
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**EDUCATION**

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| <b>B.Sc. (Hons) Software Engineering</b> · Universiti Tunku Abdul Rahman (UTAR) | 2023 |
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